NAME :D.G.RAGHAVI **COURSE CODE** : CSA1013

REG NO: 192521173 **COURSE NAME** : SOFTWARE ENGINEERING

CO4-ASSESSMENT TOOL - 01

Title: AI-Based Smart Construction Project Monitoring and Risk Assessment Platform

1. Objective

The objective of testing is to verify that the AI-based construction platform correctly monitors construction progress, detects safety violations, predicts project delays, optimizes resource allocation, tracks equipment utilization, generates project dashboards, and assesses construction risks.
Testing also ensures that the platform is **accurate, reliable, secure, usable, responsive, and capable of handling real-time construction data** from drones, cameras, IoT sensors, and BIM systems.

2. Functional Requirements to be Tested

Req. ID	Functional Requirement
FR01	User registration and login
FR02	Project creation and configuration
FR03	Add and manage construction activities
FR04	Collect data from IoT sensors
FR05	Collect construction-site data from drones/cameras
FR06	Monitor construction progress
FR07	Detect safety violations using AI
FR08	Generate safety alerts
FR09	Predict project delays
FR10	Track resource allocation and utilization
FR11	Track equipment utilization
FR12	Assess construction project risks
FR13	Generate project performance dashboards
FR14	Generate reports
FR15	Update project information
FR16	Notify managers about critical events
FR17	Support BIM/project data integration
FR18	Handle system and data errors

3. Non-Functional Requirements to be Tested

NFR ID	Requirement	Testing Focus
NFR01	Performance	Response time for monitoring and analysis
NFR02	Usability	Ease of dashboard and report usage
NFR03	Reliability	Continuous monitoring without failure
NFR04	Security	Protection of project and user data
NFR05	Compatibility	Browsers, mobile devices and sensors
NFR06	Scalability	Multiple projects/users simultaneously
NFR07	Availability	Continuous availability of monitoring system
NFR08	Accuracy	Correct progress, risk and delay predictions
NFR09	Maintainability	Easy system updates and maintenance
NFR10	Real-time processing	Timely processing of sensor/drone data

4. Test Scenarios

Scenario ID	Test Scenario
TS01	User registration and login
TS02	Create a construction project
TS03	Add project activities and schedule
TS04	Add construction site information
TS05	Receive IoT sensor data
TS06	Receive drone/camera data
TS07	Monitor construction progress
TS08	Update activity completion percentage
TS09	Detect safety violations
TS10	Generate safety alerts
TS11	Predict project delays
TS12	Track resource allocation
TS13	Monitor equipment utilization
TS14	Assess project risks
TS15	Generate project dashboard
TS16	Generate project reports

TS17	Integrate BIM data
TS18	Update project information
TS19	Handle invalid and missing data
TS20	Handle sensor/network/AI failures
TS21	Send critical notifications
TS22	Verify security and access control
TS23	Verify system performance
TS24	Verify compatibility and usability

5. Detailed Test Cases

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC01	User registration	Open registration → enter details → submit	Name: Arun, Email: arun@gmail.com, Password: Arun@123	Account is created successfully	To be executed	Not Executed
TC02	Invalid registration email	Enter invalid email → submit	arun@	Validation message is displayed	To be executed	Not Executed
TC03	Existing user registration	Register using existing email	Existing email	System rejects duplicate account	To be executed	Not Executed
TC04	Valid login	Enter valid credentials → Login	Valid email/password	User successfully logs in	To be executed	Not Executed
TC05	Invalid login	Enter incorrect password	Wrong password	Login is rejected	To be executed	Not Executed
TC06	Create project	Login → New Project → enter details → Save	Project: ABC Building	Project is created successfully	To be executed	Not Executed
TC07	Create project with missing data	Leave project name blank → Save	Blank project name	Required-field error is displayed	To be executed	Not Executed
TC08	Add project activity	Open project → Add Activity	Foundation Work	Activity is added successfully	To be executed	Not Executed
TC09	Invalid project dates	Enter end date before start date	Start: 20 Aug, End: 15 Aug	System rejects invalid schedule	To be executed	Not Executed
TC10	Update activity progress	Select activity → enter completion percentage	60%	Progress is updated to 60%	To be executed	Not Executed
TC11	Progress boundary – 0%	Enter progress	0%	System accepts 0%	To be executed	Not Executed
TC12	Progress boundary – 100%	Enter progress	100%	System accepts 100% and marks activity complete	To be executed	Not Executed
TC13	Invalid progress	Enter progress above 100%	110%	System rejects value	To be executed	Not Executed
TC14	Negative progress	Enter progress below 0%	-10%	System rejects value	To be executed	Not Executed
TC15	IoT sensor integration	Connect sensor → send data	Temperature: 30°C	Sensor data is received and displayed	To be executed	Not Executed
TC16	Invalid sensor data	Send invalid sensor value	Temperature: abc	Invalid data is rejected/flagged	To be executed	Not Executed

TC17	Sensor data update	Send new sensor readings	Humidity: 65%	Latest sensor reading is displayed	To be executed	Not Executed
TC18	Drone data upload	Upload construction-site image/video	Site image	Image/video is successfully uploaded	To be executed	Not Executed
TC19	Unsupported drone file	Upload unsupported file	test.exe	System rejects unsupported file	To be executed	Not Executed
TC20	Construction progress detection	Upload site image → run AI analysis	Current site image	AI identifies construction progress	To be executed	Not Executed
TC21	Safety violation detection	Upload site image containing worker without helmet	Worker without helmet	AI detects violation and generates alert	To be executed	Not Executed
TC22	Valid safety condition	Upload image showing worker with PPE	Helmet + safety vest	No false safety violation is generated	To be executed	Not Executed
TC23	Multiple safety violations	Upload image containing multiple violations	No helmet + no vest	All detectable violations are reported	To be executed	Not Executed
TC24	Safety alert generation	Trigger safety violation	No helmet	Critical safety alert is sent to authorized manager	To be executed	Not Executed
TC25	Delay prediction	Provide delayed activity data → run prediction	Activity behind schedule	System predicts possible project delay	To be executed	Not Executed
TC26	On-time project prediction	Provide schedule data showing activities on time	Activities on schedule	System should not incorrectly report major delay	To be executed	Not Executed
TC27	Resource allocation	Add workers/equipment to activity	10 workers	Resources are assigned successfully	To be executed	Not Executed
TC28	Resource over-allocation	Assign same limited resource to conflicting activities	5 workers assigned to two activities	System identifies/flags over-allocation	To be executed	Not Executed
TC29	Equipment registration	Add construction equipment	Excavator E01	Equipment is registered	To be executed	Not Executed
TC30	Equipment utilization tracking	Record equipment usage	Excavator: 8 hours	Utilization is calculated/displayed	To be executed	Not Executed
TC31	Equipment idle tracking	Record equipment as unused	Excavator: 0 hours	System identifies idle equipment	To be executed	Not Executed
TC32	Risk assessment	Enter project conditions → assess risk	Heavy rain + delayed activity	System identifies relevant project risks	To be executed	Not Executed
TC33	Low-risk project	Enter normal project conditions	Normal weather + on-time work	Low/normal risk is displayed	To be executed	Not Executed
TC34	High-risk condition	Enter multiple critical risks	Unsafe work + schedule delay	High/critical risk is generated	To be executed	Not Executed
TC35	Dashboard generation	Open project dashboard	Project ABC	Progress, risks, resources and equipment information is displayed	To be executed	Not Executed
TC36	Dashboard update	Update activity progress → refresh dashboard	40% → 70%	Dashboard reflects latest progress	To be executed	Not Executed
TC37	Report generation	Select project → Generate Report	Monthly report	Report is generated successfully	To be executed	Not Executed

TC38	Empty report	Generate report for project without data	No project records	Appropriate no-data message is displayed	To be executed	Not Executed
TC39	BIM integration	Upload/import BIM project data	Valid BIM file	BIM information is imported correctly	To be executed	Not Executed
TC40	Invalid BIM file	Upload invalid BIM data	Corrupted file	System rejects file and displays error	To be executed	Not Executed
TC41	Update project	Change project information → Save	Updated completion date	New information is stored	To be executed	Not Executed
TC42	Critical notification	Trigger critical risk/safety violation	Critical risk	Authorized manager receives notification	To be executed	Not Executed
TC43	Network failure	Disconnect network while receiving sensor data	Network unavailable	System displays connection error and recovers without data corruption	To be executed	Not Executed
TC44	AI service failure	Request AI analysis when AI service is unavailable	Valid site image	Error message is displayed and system remains stable	To be executed	Not Executed
TC45	Unauthorized access	User attempts to access another project's data	Unauthorized project ID	Access is denied	To be executed	Not Executed
TC46	Logout security	Logout → press browser Back	Valid session	Protected project information is not accessible	To be executed	Not Executed
TC47	Performance	Upload site image and request analysis	Standard image	Analysis completes within specified response time	To be executed	Not Executed
TC48	Multiple users	Multiple project managers access dashboard simultaneously	Multiple users	System remains stable and responsive	To be executed	Not Executed
TC49	Mobile compatibility	Open dashboard on mobile browser	Android/iOS	Dashboard is usable and properly displayed	To be executed	Not Executed
TC50	Real-time monitoring	Send sensor data → observe dashboard	New sensor reading	Dashboard updates within specified real-time interval	To be executed	Not Executed

6. Equivalence Partitioning

Equivalence Partitioning divides input values into valid and invalid groups.

Example: Project Progress

Partition	Test Data	Expected Result
Valid	0–100%	Accepted
Invalid – below range	Less than 0%	Rejected
Invalid – above range	Greater than 100%	Rejected
Invalid – non-numeric	abc	Rejected
Empty	Blank	Validation message

Example: Equipment Utilization

Partition	Test Data	Expected Result
Valid	0–24 hours/day	Accepted
Invalid – negative	-5 hours	Rejected
Invalid – above daily limit	30 hours	Rejected
Non-numeric	abc	Rejected
Empty	Blank	Required-field message

8. Boundary Value Analysis

Boundary Value Analysis tests values immediately below, at, and above acceptable boundaries.

Project Progress

Boundary	Test Value	Expected Result
Minimum − 1	-1%	Reject
Minimum	0%	Accept
Minimum + 1	1%	Accept
Maximum − 1	99%	Accept
Maximum	100%	Accept
Maximum + 1	101%	Reject

Equipment Usage

Assuming the maximum valid recorded utilization is **24 hours/day**:

Boundary	Test Value	Expected Result
Minimum − 1	-1 hr	Reject
Minimum	0 hr	Accept
Minimum + 1	1 hr	Accept
Maximum − 1	23 hr	Accept
Maximum	24 hr	Accept
Maximum + 1	25 hr	Reject

9. Decision Table Testing

Scenario: Construction Risk Assessment

Conditions:

- **C1:** Safety violation detected
- **C2:** Project activity delayed
- **C3:** Equipment/resource problem
- **C4:** Severe weather/environmental condition

Rule	C1 Safety Violation	C2 Delay	C3 Resource Problem	C4 Severe Weather	Expected Result

R1	Yes	Yes	Yes	Yes	Critical project risk
R2	Yes	Yes	No	No	High risk + safety alert
R3	Yes	No	No	No	Safety risk + alert
R4	No	Yes	Yes	No	High schedule/resource risk
R5	No	Yes	No	No	Schedule delay risk
R6	No	No	Yes	No	Resource utilization risk
R7	No	No	No	Yes	Environmental/weather risk
R8	No	No	No	No	Normal/low risk

Example:
If the system detects a **safety violation + project delay + equipment problem + severe weather**, the system should classify the situation as a **critical project risk** and notify the responsible manager.

10. State Transition Testing
The construction monitoring platform can be tested using different project states.

Project State Flow

Project Created → Planned → In Progress → Delayed/At Risk → Completed

Current State	**Event**	**Next State**	**Expected Result**
No Project	Create project	Project Created	Project is created
Project Created	Add schedule/resources	Planned	Project becomes ready for execution
Planned	Start construction	In Progress	Monitoring begins
In Progress	Normal progress	In Progress	Monitoring continues
In Progress	Delay detected	Delayed/At Risk	Risk/delay alert generated
In Progress	Safety violation	At Risk	Safety alert generated
Delayed/At Risk	Corrective action	In Progress	Risk status updated
In Progress	All activities completed	Completed	Project marked completed
Completed	View dashboard	Completed	Final project information displayed

11. Non-Functional Test Cases

TC ID	Test Area	Test Steps	Expected Result
NFT01	Performance	Upload site image and request AI analysis	Result is generated within specified response time
NFT02	Real-time processing	Send IoT sensor reading	Dashboard reflects the latest reading within the specified interval
NFT03	Reliability	Run continuous monitoring for an extended period	System continues operating without unexpected failure
NFT04	Security	Attempt unauthorized project access	Access is denied
NFT05	Security	Logout and press Back	Protected project information remains inaccessible
NFT06	Compatibility	Test Chrome, Edge and Firefox	Core functions work correctly
NFT07	Mobile usability	Open dashboard on smartphone	Dashboard is readable and usable
NFT08	Scalability	Simulate multiple users/projects	System remains stable
NFT09	Data accuracy	Compare calculated progress with source data	Correct progress percentage is displayed
NFT10	AI accuracy	Test known safety-violation images	Expected violations are detected
NFT11	Availability	Access system during normal operating hours	Platform remains available
NFT12	Recovery	Disconnect network and reconnect	System recovers without corrupting project data

12. Traceability Matrix

Requirement	Related Scenarios	Test Cases
FR01 Registration/Login	TS01	TC01–TC05
FR02 Project Management	TS02	TC06–TC07
FR03 Construction Activities	TS03, TS08	TC08–TC14
FR04 IoT Data	TS05	TC15–TC17
FR05 Drone/Camera Data	TS06	TC18–TC20
FR06 Progress Monitoring	TS07	TC20, TC35–TC36

FR07 Safety Detection	TS09	TC21–TC23
FR08 Safety Alerts	TS10	TC24, TC42
FR09 Delay Prediction	TS11	TC25–TC26
FR10 Resource Allocation	TS12	TC27–TC28
FR11 Equipment Tracking	TS13	TC29–TC31
FR12 Risk Assessment	TS14	TC32–TC34
FR13 Dashboard	TS15	TC35–TC36
FR14 Reports	TS16	TC37–TC38
FR15 Project Updates	TS18	TC41
FR16 Notifications	TS21	TC24, TC42
FR17 BIM Integration	TS17	TC39–TC40
FR18 Error Handling	TS19, TS20	TC43–TC44
NFR01 Performance	TS23	TC47
NFR02 Usabilit	TS24	TC49
NFR03 Reliability	TS23	NFT03
NFR04 Security	TS22	TC45–TC46, NFT04–NFT05
NFR05 Compatibility	TS24	NFT06–NFT07
NFR06 Scalability	TS23	TC48, NFT08
NFR08 Accuracy	TS23	NFT09–NFT10
NFR10 Real-time Processing	TS23	TC50, NFT02
Requirement	**Related Scenarios**	**Test Cases**
FR01 Registration/Login	TS01	TC01–TC05

13. Test Coverage Summary

Testing Area	**Coverage**
Functional Requirements	✓ High
Non-Functional Requirements	✓ High
Normal Conditions	✓
Boundary Conditions	✓
Invalid Conditions	✓
Exceptional Conditions	✓
Equivalence Partitioning	✓
Boundary Value Analysis	✓
Decision Table Testing	✓
State Transition Testing	✓
AI/Safety Detection	✓
Risk Assessment	✓
Security Testing	✓
Performance Testing	✓
Compatibility Testing	✓

Real-Time Monitoring	✓
Traceability	✓

Requirement Coverage Formula

Requirement Coverage = (Requirements covered by test cases / Total requirements) × 100
For example, if all 18 functional requirements are covered:
Requirement Coverage = (18 / 18) × 100 = 100%

14. Overall Test Result / Conclusion

The proposed test case design provides comprehensive testing for the **AI-Based Smart Construction Project Monitoring and Risk Assessment Platform**. The test cases cover construction project creation, activity monitoring, IoT sensor integration, drone/camera data processing, AI-based safety violation detection, delay prediction, resource allocation, equipment utilization, risk assessment, dashboards, reporting, BIM integration, notifications, security, performance, and real-time monitoring.
The test design uses **Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing** to cover normal, boundary, invalid, and exceptional conditions.
The traceability matrix establishes a clear relationship between **requirements, test scenarios, and test cases**, providing strong requirement and functional coverage.